

0 8 . 1

The letters **A**, **B**, **C**, **D** and **E** represent stages in mitosis.

- **A** – anaphase
- **B** – interphase
- **C** – metaphase
- **D** – prophase
- **E** – telophase

Write **one** of the letters, **A** to **E**, in the box to complete the following statement.

[1 mark]

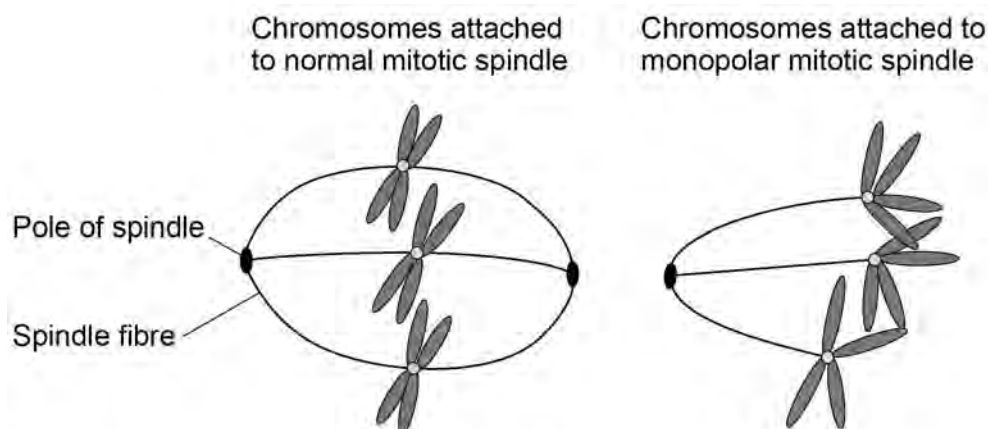
Chromosomes line up on the equator of the mitotic spindle in

0 8 . 2

Scientists looking for treatments for cancer are investigating the use of substances called kinesin inhibitors (KI). These inhibitors prevent successful mitosis. Some kinesin inhibitors cause the development of a monopolar spindle in mitosis.

**Figure 6** shows chromosomes attached to a normal mitotic spindle and to a monopolar mitotic spindle.

**Figure 6**



Suggest why the development of a monopolar mitotic spindle would prevent successful mitosis.

[2 marks]

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Scientists investigated the effect of different concentrations of a kinesin inhibitor (KI) on mitosis of human bone-cancer cells grown in a culture.

**Table 3** shows the scientists' results.

### Table 3

| Concentration of kinesin inhibitor /<br>nmol dm <sup>-3</sup> | Percentage of dividing human<br>bone-cancer cells showing a<br>monopolar mitotic spindle |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 0                                                             | 0                                                                                        |
| 1                                                             | 0                                                                                        |
| 10                                                            | 8                                                                                        |
| 100                                                           | 93                                                                                       |
| 1000                                                          | 100                                                                                      |
| 10 000                                                        | 100                                                                                      |

A student who saw these results concluded that in any future trials of this kinesin inhibitor with people, a concentration of  $100 \text{ nmol dm}^{-3}$  would be most appropriate to use.

Do these data support the student's conclusion? Give reasons for your answer. **[4 marks]**

[illegible]

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At the start of their investigation, the scientists made a solution of kinesin inhibitor (KI) with a concentration of  $10\,000\text{ nmol dm}^{-3}$ . They used this to make the other concentrations by a series of dilutions with water.

Describe how they made  $100\text{ cm}^3$  of  $1000\text{ nmol dm}^{-3}$  solution of kinesin inhibitor.  
**[2 marks]**

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**Turn over for the next question**

